

Experiment 5

Connecting Devices to a Wi-Fi Network

Lab 5 Objectives:

- Connecting Devices to a Wi-Fi Network, Implementing WPA2 Security, and Testing Connectivity

Prerequisites: The prerequisites of this lab are-

- Basic knowledge of networking concepts and wireless communication.
- Familiarity with Wi-Fi security protocols and encryption standards.
- Access to a router with WPA2 security and wireless-enabled devices.

Outcomes: By the end of this lab, students will be able to:

- Configure a Wi-Fi network and connect different devices securely.
- Implement WPA2 encryption for wireless security.
- Test network connectivity and troubleshoot common Wi-Fi issues.

Part 1: Understanding Wi-Fi Networks and Security

Step 1: Wi-Fi Fundamentals

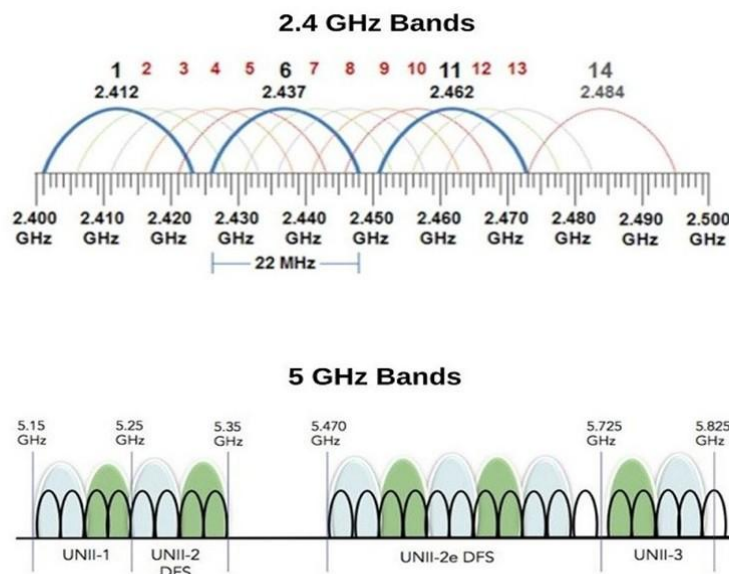
Overview of Wi-Fi Standards (802.11a/b/g/n/ac/ax):

Wi-Fi has various standards that define speed, range, and frequency:

- **802.11a:** Uses 5 GHz band, speeds up to 54 Mbps, less interference.
- **802.11b:** Uses 2.4 GHz band, speeds up to 11 Mbps, more interference.
- **802.11g:** Uses 2.4 GHz band, speeds up to 54 Mbps, backward compatible with 802.11b.
- **802.11n:** Dual band (2.4 & 5 GHz), speeds up to 600 Mbps, uses MIMO technology.
- **802.11ac:** Uses 5 GHz band, speeds up to several Gbps, faster and better range.
- **802.11ax (Wi-Fi 6):** Latest standard with improved efficiency, speed, and capacity, supports both 2.4 & 5 GHz.

Understanding SSID, Channels, Frequency Bands (2.4 GHz vs 5 GHz):

- **SSID (Service Set Identifier):** The unique name of a Wi-Fi network.
- **Channels:** Wi-Fi operates on specific channels within frequency bands. The 2.4 GHz band has 1-11 channels, often overlapping; 5 GHz has more non-overlapping channels.
- **Frequency Bands:**
 - **2.4 GHz:** Longer range but slower speeds and more interference.
 - **5 GHz:** Faster speeds, less interference, but shorter range.



Step 2: Wi-Fi Security Basics

Explanation of WEP, WPA, WPA2, and WPA3 Security Protocols:

- **WEP (Wired Equivalent Privacy):** An outdated and insecure protocol.
- **WPA (Wi-Fi Protected Access):** Improved security over WEP but still vulnerable.
- **WPA2:** Currently the standard security protocol using AES encryption, highly secure.
- **WPA3:** The newest security standard with enhanced protection against attacks.

Importance of Encryption:

Encryption protects wireless data from unauthorized access and ensures privacy by encoding the data transmitted over the network.

	WEP	WPA	WPA2	WPA3
Release Year	1999	2003	2004	2018
Encryption Method	Rivest Cipher 4 (RC4)	Temporal Key Integrity Protocol(TKIP) with RC4	CCMP and Advanced Encryption Standard	Advanced Encryption Standard(AES)
Session Key Size	40-bit	128-bit	128-bit	128-bit(WPA3-Personal) 192-bit(WPA3-Enterprise)
Cipher Type	Stream	Stream	Block	Block
Data Integrity	CRC-32	Message Integrity Code	CBC-MAC	Secure Hash Algorithm
Key Management	Not provided	4-way handshaking mechanism	4-way handshaking mechanism	Simultaneous Authentication of Equals handshark
Authentication	WPE-Open WPE-Shared	Pre-Shared Key(PSK)& 802.1x with EAP variant	Pre-Shared Key(PSK)& 802.1x with EAP variant	Simultaneous Authentication of Equals(SAE)&802.1x with EAP variant

Part 2: Configuring and Securing a Wi-Fi Network

Step 1: Setting Up a Wi-Fi Network

- Access the router's admin panel via a web browser by typing the router's IP address (commonly 192.168.1.1 or 192.168.0.1).
- Login with the admin username and password.
- Set the **SSID** (network name).
- Select the network mode (such as 802.11n/ac/ax).
- Set a secure password for the Wi-Fi network.

The screenshot shows the WatchGuard configuration interface. On the left is a dark sidebar with navigation links: DASHBOARD, MONITOR, CONFIGURE (highlighted in red), TROUBLESHOOT, FLOOR PLANS, REPORTS, and SYSTEM. The main content area is titled 'WiFi' with a dropdown arrow and a 'SSID' tab. Below this is a 'SSID Name' section with a back arrow icon and tabs for 'Basic', 'Security', 'Network', and 'SSID'. The 'Basic' tab is active. It contains two text input fields: 'SSID Name *' with a placeholder 'Enter SSID Name' and 'Profile Name *' with a placeholder 'Enter Profile Name'. Below these is a 'Select SSID Type' section with two radio buttons: 'Private' (selected) and 'Guest'. At the bottom are two checkboxes: 'Hide SSID' and 'Include AP Name in Beacon'.

Step 2: Implementing WPA2 Security

- Enable **WPA2-Personal** security mode in router settings.
- Choose **AES** as the encryption method for strong security.
- Generate and set a strong passphrase containing uppercase, lowercase letters, numbers, and special characters.
- Save the settings.

Example: **123456780** ❌ **Akdfudfni*@7990** ✅

Step 3: Connecting Devices to the Network

- On the laptop, smartphone, and IoT devices, open Wi-Fi settings.
- Select the SSID configured earlier.
- Enter the WPA2 password and connect.
- Verify devices are connected using WPA2 encryption.

Part 3: Testing and Analyzing Network Connectivity

Step 1: Checking Connectivity

- Use ping command in Command Prompt or Terminal to test internet connectivity, e.g.

```
C:\WINDOWS\system32\cmd. x + v
(c) Microsoft Corporation. All rights reserved.
C:\Users\CHETAN>ping google.com

Pinging google.com [142.250.192.142] with 32 bytes of data:
Reply from 142.250.192.142: bytes=32 time=56ms TTL=113
Reply from 142.250.192.142: bytes=32 time=59ms TTL=113
Reply from 142.250.192.142: bytes=32 time=78ms TTL=113
Reply from 142.250.192.142: bytes=32 time=43ms TTL=113

Ping statistics for 142.250.192.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 78ms, Average = 59ms
```

- Use tracert (Windows) or traceroute (Linux/macOS) to view the path data takes to reach a website:

```
C:\Users\CHETAN>tracert google.com

Tracing route to google.com [142.250.192.142]
over a maximum of 30 hops:

  0  9 ms    3 ms    3 ms   192.168.226.177
  1 193 ms   273 ms  127 ms  192.168.29.10
  2  59 ms    35 ms    39 ms  192.168.28.205
  3  68 ms    27 ms    33 ms  192.168.31.2
  4  42 ms    37 ms    33 ms  192.168.31.33
  5  44 ms    27 ms    36 ms  192.168.251.98
  6  59 ms    39 ms    33 ms  125.20.115.138
  7  57 ms    25 ms    41 ms  125.20.115.137
  8  62 ms    52 ms    67 ms  116.119.36.184
  9  61 ms    50 ms    59 ms  72.14.213.254
 10 85 ms   203 ms    56 ms  172.253.69.227
 11 222 ms    52 ms    49 ms  142.250.238.81
 12  61 ms    47 ms    62 ms  bom12s18-in-f14.1e100.net [142.250.192.142]

Trace complete.
C:\Users\CHETAN>
```

- Check both internet and local network connectivity.